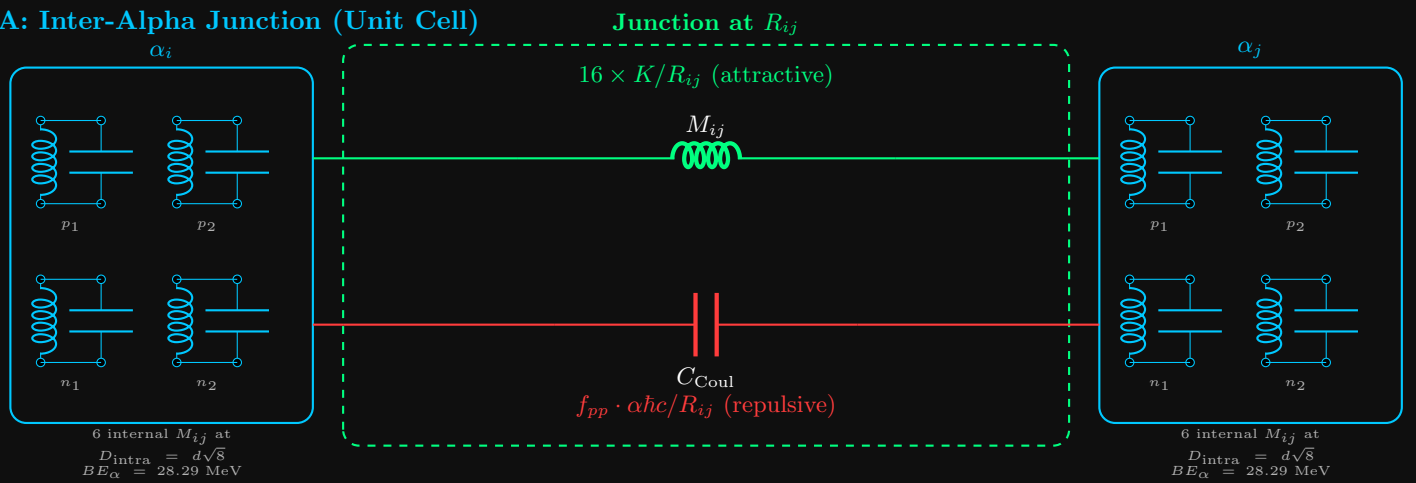
# Ca-40 Semiconductor Equivalent Circuit

$10\alpha$ Bicapped Antiprism — $V_R/V_{BR} = 0.994$ — $M = 32.900$ (Large Signal)

## A: Inter-Alpha Junction (Unit Cell)



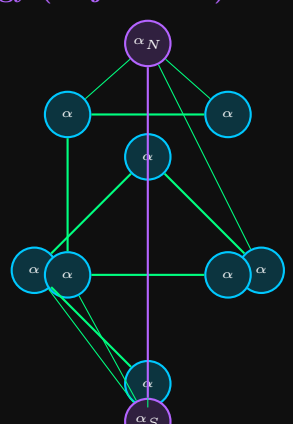

## B: Miller Avalanche Stage

$V_{BR} = 6\alpha\hbar c/D_{\text{intra}} = 3.594$ MeV

$$M = \frac{1}{1 - \left(\frac{V_R}{V_{BR}}\right)^5}$$

Ca-40: $V_R/V_{BR} = 0.994 \implies M = 32.9$ (**AVALANCHE**)

## C: Ca-40 Topology (45 junctions)




8 ring + 6 cap
+31 cross-links
**45 junctions**

## D: Energy Balance

$$M_{\text{nuc}} = 10\,M_\alpha - BE_{\text{net}}$$

$$BE_{\text{net}} = \underbrace{\sum_{i<j}^{45} 16 \cdot \frac{K}{R_{ij}}}_{textStrong(attractive)} - \underbrace{M \cdot \sum f_{pp}\,\frac{\alpha\hbar c}{R_{ij}}}_{\text{Coulomb (repulsive)}}$$

Strong: $\sum K/R =$ engine-derived
Coulomb: $\sum \alpha\hbar c/R \times f_{pp}$
Miller: $M = 32.900$ ($V_R/V_{BR} = 0.994$)

**Result:** $37,322.573$ MeV (0.0000% error)

Each $\alpha = 2p + 2n$ LC tanks at tetrahedron vertices ($D_{\text{intra}} = d\sqrt{8} \approx 2.38$ fm).
$d = 4\hbar/m_p c \approx 0.841$ fm. $K_{\text{mutual}} = 11.34$ MeV·fm. $V_{BR} = 3.594$ MeV.